

AUTOSAR Engine Control Unit (ECU) High-Level Design

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1. Executive Architecture Overview

This document describes the AUTOSAR Classic architecture for the Engine Control Unit (ECU). The software system follows AUTOSAR 4.4 specifications, separating Application Software Components (SWCs) from the Basic Software (BSW) stack via the Runtime Environment (RTE).

2. Software Component (SWC) Specification

The application layer contains two primary components: **EngineSpeedControl_SWC** and **ThrottleControl_SWC**. EngineSpeedControl_SWC receives crankshaft sensor frequency data and calculates required torque output. ThrottleControl_SWC commands the throttle actuator position.

SWC Name	Component Type	Provided Ports	Required Ports
EngineSpeedControl_SWC	Atomic Application SWC	PPort_EngineSpeed	RPort_ThrottlePos
ThrottleControl_SWC	Atomic Application SWC	PPort_ThrottleActuator	RPort_EngineSpeed

3. Basic Software (BSW) Stack Configuration

The BSW layer includes **EcuM** (ECU State Manager), **BswM** (BSW Mode Manager), **CanIf** (CAN Interface), **Com** (Communication Stack), and **Dem** (Diagnostic Event Manager). CanIf handles frame routing between CanTp and MCAL CAN driver.

4. Interface & Signal Specifications

Inter-component communication relies on *SenderReceiverInterface*. Signal **Sig_EngineSpeed_rpm** transmits engine rotation velocity, while **Sig_ThrottlePos_pct** transmits percentage opening. Diagnostic errors are logged into Dem via *ClientServerInterface*.